

Ratun Rahman

PHD STUDENT IN COMPUTER ENGINEERING (QUANTUM COMPUTING)

Huntsville, AL 35816, USA

+1 (256) 479-9640 | ✉ rr0110@uah.edu | 🌐 <https://sites.google.com/uah.edu/ratunrahman> | 📧 ratun11 | 🌐
ratun-rahman-46762b187 | 🎓 Ratun Rahman

Profile

Ph.D. student in Computer Engineering specializing in noise-resilient quantum machine learning and distributed quantum systems. My research focuses on developing error mitigation techniques, including zero-noise extrapolation and noise-aware training, to enable reliable learning on noisy intermediate-scale quantum (NISQ) devices. I design scalable quantum-classical frameworks that address instability, heterogeneity, and performance limitations in practical quantum environments.

My work aims to bridge the gap between theoretical quantum algorithms and real-world applications by improving the robustness and usability of quantum learning systems. I have applied these methods to domains such as energy forecasting, intelligent sensing, and distributed systems, with the goal of enabling dependable quantum computing for next-generation engineering applications.

Education

University of Alabama in Huntsville

Huntsville, AL, USA

PH.D. IN COMPUTER ENGINEERING (QUANTUM COMPUTING & DISTRIBUTED SYSTEMS)

Aug 2023 – Dec 2026 (Expected)

GPA: 3.83 / 4.00

Islamic University of Technology

Gazipur, Bangladesh

B.SC. IN SOFTWARE ENGINEERING (SECURITY AND MACHINE LEARNING)

Jan 2018 – May 2022

GPA: 3.62 / 4.00

Technical Expertise

QUANTUM COMPUTING & QML: Quantum Machine Learning, Variational Quantum Circuits, Noise Mitigation (Zero-Noise Extrapolation, Dynamical Decoupling, Readout Error Mitigation), Noise-Aware Training

DISTRIBUTED & LEARNING SYSTEMS: Federated Learning, Distributed Quantum Learning, Model Personalization, Heterogeneous Systems

FRAMEWORKS & TOOLS: Qiskit, Qiskit Aer, PennyLane

PROGRAMMING: Python, C++, Rust, PyTorch, CUDA, NumPy

SYSTEMS & COMPUTING: HPC Workflows, Distributed Execution, Experiment Orchestration, GPU Acceleration, Linux

Publications

Journal Publications

1. **Ratun Rahman**, Shaba Shaon, and Dinh C. Nguyen. "Tackling Heterogeneity in Quantum Federated Learning: An Integrated Sporadic-Personalized Approach." *Accepted at IEEE Transactions on Computers*, 2026.
2. **Ratun Rahman**, Dinh C. Nguyen. "Quantum Convolutional Neural Networks: A Survey on Architectures, Applications, and Future Directions." *Accepted at IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, 2026.
3. **Ratun Rahman**. "Geometric Fairness-Aware Routing for Federated Edge Networks." *Accepted at IEEE Transactions on Networking*, 2026.
4. **Ratun Rahman**, Dinh C. Nguyen. "Probabilistic Federated Learning on Uncertain and Heterogeneous Data with Model Personalization." *Accepted at IEEE Transactions on Emerging Topics in Computational Intelligence (TETCI)*, 2026.
5. **Ratun Rahman**, Sina Shaham, and Dinh C. Nguyen. "Towards Personalized Quantum Federated Learning for Anomaly Detection." *Published in IEEE Transactions on Network Science and Engineering*, 2025.
6. **Ratun Rahman**, Dinh C. Nguyen, Christo Kurisummootil Thomas, and Walid Saad. "Towards Heterogeneous Quantum Federated Learning: Challenges and Solutions." *Published in IEEE Network*, 2025.
7. **Ratun Rahman** and Dinh C. Nguyen. "Escaping Barren Plateaus in Variational Quantum Algorithms Using Negative Learning Rate." *Published in IEEE Internet of Things Journal*, 2025.
8. **Ratun Rahman**, Pablo Moriano, Samee U. Khan, and Dinh C. Nguyen. "Electrical Load Forecasting over Multihop Smart Metering Networks with Federated Learning." *Published in IEEE Internet of Things Journal*, 2025.
9. **Ratun Rahman**, Dinh C. Nguyen. "Improved Modulation Recognition Using Personalized Federated Learning." *Published in IEEE Transactions on Vehicular Technology*, 2024.
10. **Ratun Rahman**, F. Patadia, J. Wang, and Dinh C. Nguyen. "Multimodal Federated Learning for Air Quality Estimation with Model Personalization over Aerial-Ground Networks." *Published in IEEE Geoscience and Remote Sensing Letters*, 2025.

Conference Publications

1. **Ratun Rahman**, Dinh C. Nguyen. “Multimodal Federated Learning with Model Personalization.” *OPT, Presented at NeurIPS*, 2024.
2. **Ratun Rahman**, Atit Pokharel, and Dinh C. Nguyen. “Toward High-Precision Variational Quantum Sensing via Asynchronous Federated Learning.” *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2026.
3. Atit Pokharel, **Ratun Rahman**, Thomas Morris, and Dinh C. Nguyen. “Differential Privacy for Distributed Quantum Sensing via Quantum Noise.” *Published at IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2026.
4. **Ratun Rahman**, Atit Pokharel, and Dinh C. Nguyen. “SimQFL: A Quantum Federated Learning Simulator with Real-Time Visualization.” *Published at IEEE QCE*, 2025.
5. Atit Pokharel, **Ratun Rahman**, Shaba Shaon, Thomas Morris, and Dinh C. Nguyen. “Differentially Private Federated Quantum Learning via Quantum Noise.” *Published at IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2025.
6. **Ratun Rahman**, Atit Pokharel, and Dinh C. Nguyen. “Sporadic Federated Learning in Quantum Environments to Tackle Noise.” *FedVision, Presented at CVPR*, 2025.
7. Atit Pokharel, **Ratun Rahman**, Thomas Morris, and Dinh C. Nguyen. “Quantum Federated Learning for Multimodal Data: A Modality-Agnostic Approach.” *BEAM, Presented at CVPR*, 2025.
8. **Ratun Rahman**, Dinh C. Nguyen. “Electrical Load Forecasting in Smart Grid: A Personalized Federated Learning Approach.” *Published at IEEE CCNC*, 2024.

Projects

1. **QuSenseSim: Interactive Quantum Sensing Simulation Framework** (2026) (<https://github.com/Ratun11/QuSenseSim>)
Developed a modular and interactive simulation framework for multi-sensor quantum sensing systems under realistic and heterogeneous noise environments. Integrated parameterized quantum circuit modeling with adaptive noise injection and personalized mitigation strategies, enabling real-time evaluation using Fisher Information (FI) and Cramér–Rao Bound (CRB). Supports multi-sensor configurations with user-controlled noise, mitigation, and collaboration settings, facilitating system-level analysis of sensing performance and robustness in NISQ conditions.
2. **QuNoise: Quantum Noise Simulator and Visual Analytics Platform** (2026) (<https://github.com/Ratun11/QuNoise>)
Developed a noise-aware simulation and visualization framework to analyze the impact of depolarization, amplitude damping, phase damping, and readout errors on quantum learning systems. Implemented advanced error mitigation techniques including zero-noise extrapolation (ZNE), probabilistic error cancellation (PEC), and measurement error mitigation (MEM), enabling detailed evaluation of robustness and model stability under realistic NISQ conditions.
3. **SimQFL: Distributed Quantum Federated Learning Simulator** (2025) <https://github.com/Ratun11/SimQFL>
Designed a scalable client–server framework for heterogeneous quantum federated learning with noise-aware training. Incorporated QPU-level noise models, client heterogeneity, and secure aggregation to support large-scale distributed quantum learning experiments.
4. **ZNE-Rigetti: Real-Hardware Quantum Noise Mitigation on Rigetti QPU** (2026) (<https://github.com/Ratun11/zne-rigetti-real-quantum-device>)
Developed a real-hardware quantum noise mitigation framework using Amazon Braket and the Rigetti Cepheus-1-108Q superconducting quantum processor. Implemented zero-noise extrapolation (ZNE) with multiple folding-scale arms, task-level QPU execution tracking, Pauli- Z expectation-value estimation, and cost-aware arm-wise evaluation. Conducted real-device validation using CIFAR-10 quantum inference samples with 240 completed QPU tasks, enabling practical analysis of accuracy, loss, entropy, reward, runtime, and hardware cost under real NISQ execution constraints.

Work Experience

Graduate Research Assistant — Network Intelligence and Security Lab

Quantum Computing and Distributed Systems

University of Alabama in Huntsville (UAH), Huntsville, AL

Aug 2023 – Present

- Developed hybrid quantum–classical learning pipelines for noisy intermediate-scale quantum (NISQ) devices using Qiskit and PennyLane.
- Designed noise-resilient quantum learning frameworks incorporating error mitigation techniques such as Zero-Noise Extrapolation (ZNE), measurement error mitigation, and probabilistic error cancellation.
- Built scalable distributed quantum learning systems with client-level heterogeneity and QPU-specific noise modeling.
- Implemented benchmarking and orchestration tools to evaluate convergence, fidelity, and robustness across simulated quantum backends.
- Optimized large-scale quantum simulations using HPC-style parallel execution and GPU acceleration.

Honors & Awards

2026-2027	Alabama EPSCoR GRSP Round 21 Fellowship , Received from the Alabama Commission for Higher Education (ACHE).	<i>UAH, USA</i>
2026	Mary Makima and Lester Ross Scholarship , Received from the College of Engineering Scholarship for graduate students.	<i>UAH, USA</i>
2025	Featured Research , Oak Ridge National Laboratory highlighted my work on federated learning for smart metering networks. Access here (https://www.ornl.gov/publication/electrical-load-forecasting-over-multihop-smart-metering-networks-federated-learning).	<i>ORNL, USA</i>
2025	Oral Presentation , "Sporadic Federated Learning Approach in Quantum Environment to Tackle Quantum Noise" is accepted as an oral presentation in CVPR 2025	<i>CVPR, USA</i>
2025	Featured Research , Pacific Northwest National Laboratory highlighted my work on personalized federated learning for smart grid. Access here (https://www.pnnl.gov/publications/electrical-load-forecasting-smart-grid-personalized-federated-learning-approach).	<i>PNNL, USA</i>
2023	Top-30 Paper Award , Von Braun Space Exploration Symposium for work on satellite federated learning arranged by American Astronautical Society (AAS).	<i>USA</i>
2020	Finalist , ICPC International Collegiate Programming Contest selection in Bangladesh	<i>Bangladesh</i>
2019	Bronze Prize , IUT Programming Competition	<i>Bangladesh</i>